

Assessing the Effectiveness of a Constructed Arctic Stream Using Multiple Biological Attributes

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Abstract Objective assessment of habitat compensation is a central yet challenging issue for restoration ecologists. In 1997, a 3.4-km stream channel, designed to divert water around an open pit diamond mine, was excavated in the Barrenlands region of the Canadian Arctic to create productive stream habitat. We evaluated the initial success of this compensation program by comparing multiple biological attributes of the constructed stream during its first three years to those of natural reference streams in the area. The riparian zone of the constructed stream was largely devoid of vegetation throughout the period, in contrast to the densely vegetated zones of reference streams. The constructed stream also contained lower amounts of woody debris, coarse particulate organic matter (CPOM), and epilithon; had lower coverage by macrophytes and bryophytes; and processed leaf litter at a lower rate than reference streams. Species richness and densities of macroinvertebrates were consistently lower in the constructed stream compared to natural streams. This contributed to differences in macroinvertebrate assemblage structure throughout the period, although assemblages showed some convergence by year 3. The effectiveness of the constructed stream to emulate natural streams varied

somewhat depending on the biological attribute being evaluated. Assessments based on individual attributes showed that minimal to moderate levels of similarity between the constructed stream and natural streams were achieved. A collective assessment of all biological and ecosystem attributes suggested that the constructed stream was not a good surrogate for natural streams during these first years. Additional time would be required before many characteristics of the constructed stream would resemble those of reference streams. Because initial efforts to improve fish habitat in the constructed stream focused on physical structures (e.g., weirs, vanes, rock, groins), ecological factors limiting fish growth were not considered and likely constrained success. We suggest that a greater focus on organic characteristics and vegetation within the stream and its riparian zone could have accelerated compensation. The addition of woody debris and CPOM, combined with planting of shrubs and herbs along the stream, should provide a source of allochthonous matter for the biotic community while large cobble and boulders should improve the physical stability of stream system, protecting its organic components.

Keywords Productive fish habitat · No net loss · Constructed streams · Arctic · Benthic macroinvertebrates · Stream assessment

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Introduction

Quantifying what constitutes effective habitat is a central issue for programs dealing with habitat compensation or restoration, including those operating under Canada's *Fisheries Act*. Under this Act, an authorization for development that results in harmful alteration, disruption, or

destruction (HADD) of fish habitat requires compensation to offset the HADD. To guide implementation of the Act, the Department of Fisheries and Oceans (DFO) developed a “Policy for the Management of Fish Habitat” (hereafter, the Policy, DFO 1986). The guiding principle of the Policy is to ensure “no net loss of productive capacity of fish habitats” (Minns and others 1996), defined as “the maximum natural capability of habitats to produce healthy fish ... or to support or produce aquatic organisms upon which fish depend” (DFO 1986, p. 28). A recent critical evaluation of fish habitat compensation measures in Canada showed that the creation of “in-kind” (i.e., similar) habitat was the most frequently adopted option to ensure the required “no net loss” (Harper and Quigley 2005). Despite the overwhelming importance of habitat compensation measures to ensure the maintenance of fish habitat in Canada and elsewhere, surprisingly few studies have evaluated the ecological effectiveness of mitigation measures or have devised monitoring programs capable of determining success or failure (Reeves and others 1991; but see Hunt 1988; Jones and others 2003b; Harper and Quigley 2005). As a result, the effectiveness of habitat compensation programs has been broadly challenged (e.g., Minns and others 1996; Harper and Quigley 2005). While many acknowledge ecological restoration as an emerging science, our ability to ensure that habitat compensation efforts are achieving their ecological and regulatory objectives requires that compensation efforts are evaluated using a suite of ecological measures. For instance, the ecological effectiveness of in-kind habitats to offset HADD could be determined by comparing structural and functional attributes of recently constructed habitats to appropriate reference systems (Jones and others 2003b).

In 1991, diamonds were discovered in the remote region of the Northwest Territories of Canada known as the Barrenlands. In preparation for mineral extraction, two lakes and their connecting streams were drained to provide access to the diamond-bearing ore. As part of the habitat compensation agreement, a 3.4-km stream channel was blasted out of the granitic bedrock to reconnect the drainage network. Since 1997, water has been diverted around the two lakes, which became open-pit mines, through the constructed stream channel. An objective of these construction efforts was to replace the fish habitat lost during mine development.

In the present study we build on our previous work (Jones and others 2003b) that focused, in a hierarchical framework, on the ability of the constructed stream to produce fish, especially young Arctic grayling. We found that adult grayling successfully migrated through the channel and its culverts, spawned in the channel, and that eggs hatched and young grayling were produced. Despite these successes, the poor growth and production of

young-of-year (YOY) grayling strongly suggested that the quality of nursery habitat was deficient in some manner, e.g., food production, physical habitat, or water temperature (Jones and others 2003b). Although we examined the benthic invertebrates and other biological attributes in our original study, there was little detail beyond coarse composition of the assemblages or overall biomass.

Aquatic invertebrates constitute important links in lotic food webs (e.g., Poff and Huryn 1997) and are sensitive indicators of water quality and associated ecosystem services (e.g., Karr 1999). In this article, we expand our assessment of the constructed stream as “in-kind” habitat for the natural streams that were lost; in particular, we (1) evaluate the extent to which structural and functional attributes of the constructed stream’s ecosystem, including rates of organic-matter processing, approximate those of reference streams, (2) compare the structure of macroinvertebrate assemblages between the constructed and reference streams, and (3) determine whether macroinvertebrate assemblages in the constructed stream were similar to reference-stream communities over the first 3 growing seasons of the constructed stream’s existence, and whether convergence trends varied by habitat in the constructed stream. We predicted marked initial differences in structural and functional attributes, but expected that differences in select ecological attributes would show at least some partial convergence between the constructed and reference streams during the study period. Data on the constructed stream’s biotic attributes were augmented with our descriptions of structural attributes to improve our understanding of potential mechanisms behind ecological differences between the constructed and reference streams.

Methods

Study Area

The 4000 km² study area is about 100 km northeast of the tree-line within the southern Arctic ecozone (Fig. 1). The mean annual temperature is −12°C, with a mean summer maximum of 27°C and a winter minimum of −54°C (Environment Canada 1991). The climate is semi-arid, with 200–300 mm precipitation annually, about 50% falling as snow. The region is underlain by granite, gneiss, and schist that form broad, sloping uplands, plateaus, and lowlands. The permafrost layer is continuous. Eskers, kames, and boulder-strewn till plains are prominent landscape features. Thin soils in upland areas support dwarf-heath and scattered low-shrub tundra, while lowland and depressional areas near lakes and streams are characterized by sedge tussock and low-tall shrub tundra (Bliss 1981).

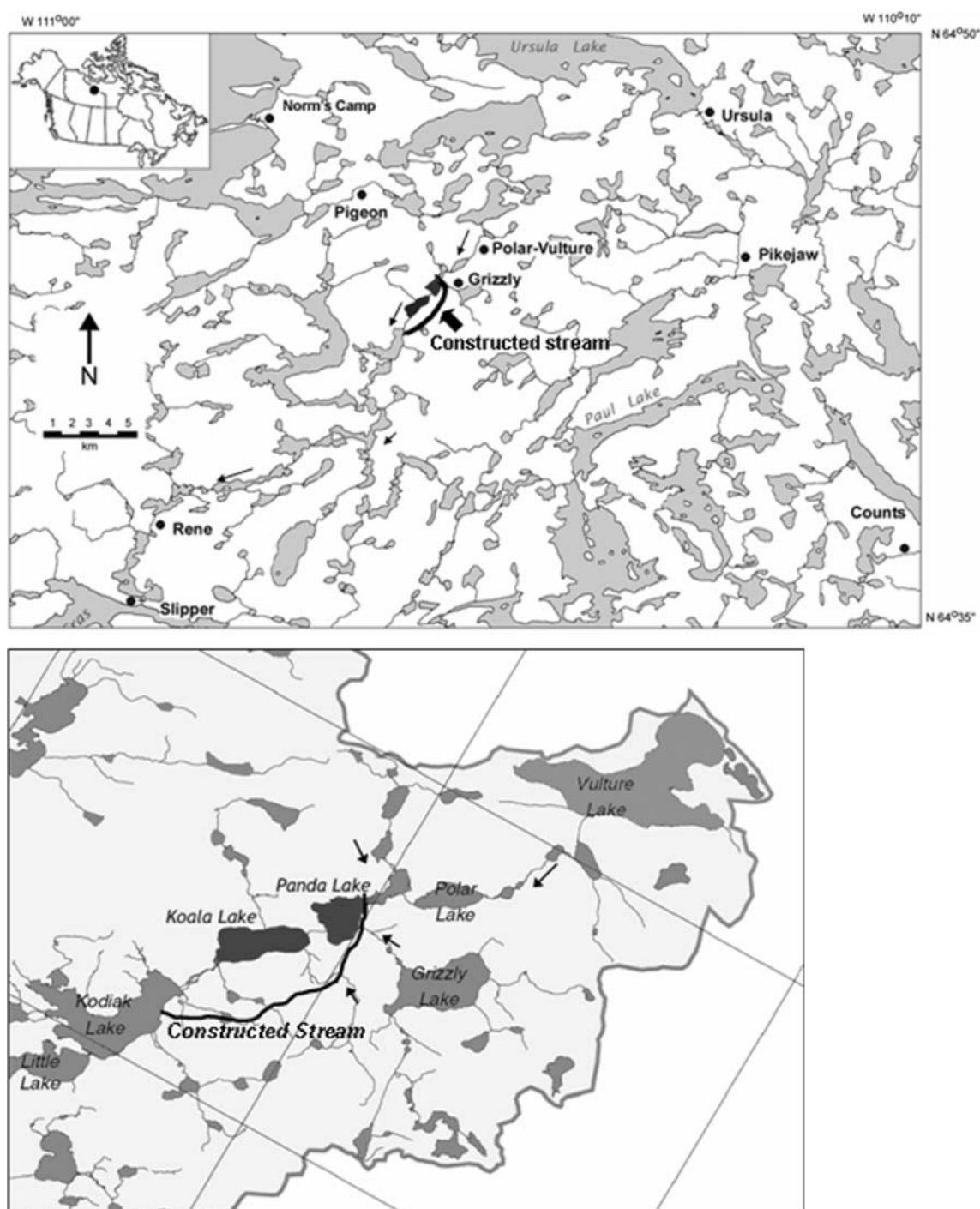


Fig. 1 Locations of the constructed stream and 10 natural streams (●) in the Barrenlands region of Northwest Territories, Canada (top). The constructed stream (thick line) is located in the centre of map. Two streams (Polar-Vulture A and Polar-Vulture B) are located in the same vicinity and are represented by a single point. Thin arrows

indicate direction of flow in the constructed stream drainage system. Smaller scale map of the constructed stream with more detail showing inflowing, largely intermittent, streams (bottom). In both maps, the darker areas (Panda and Koala lakes in the lower map) indicate de-watered basins converted to open-pit diamond mines

The combination of relatively low topographical relief (approximately 50 m), extensive glacial activity, and a shallow active layer has molded a landscape dominated by surface waters (approximately 21% of the landscape) in the

form of chains of lakes and connecting streams. Streams are generally lake outlets and travel only short distances before emptying into another lake (median length = 380 m; Jones and others 2003c). Following spring runoff,

evaporation from lakes gradually lowers lake water levels and stream flows subsequently decline, with many smaller streams characterized by intermittent discharge. Flows can increase (or resume) in late summer and fall as a result of increased precipitation, but streams freeze solid during the long arctic winter (October to June). Streams in the study area are located in the headwaters of the Coppermine, Back, and Burnside rivers, which flow north and eventually discharge to the Arctic Ocean.

Study Streams

Data were collected from the constructed stream and 10 nearby natural streams in July of 1998–2000 (i.e., the first 3 yrs following creation of the constructed stream). In each natural stream a study site was randomly chosen that was approximately four times bankfull width or at least 40 m long. Fourteen sites were sampled in the constructed stream, each varying between 60–100 m in length and were previously chosen by environmental consultants. Data from the constructed stream were pooled together for statistical testing (see Statistical Analyses), and thus, represented mean value while variability about the mean was represented in figures as standard errors. Given their remote Arctic location, natural streams were assumed to offer a representative range of undisturbed conditions for this region. As such, we used these natural streams as references against which the constructed stream could be compared. We quantified biotic and abiotic attributes of each stream and adjacent riparian zone, with separate measurements in pool and riffle habitats. We also quantified water depths and water velocities at individual locations where samples of epilithon, coarse particulate organic matter (CPOM) and macroinvertebrates were collected.

Characterizing Streams and Substratum Characteristics

We surveyed constructed and natural streams annually to characterize streams and habitats within the streams. Using ground surveys, and interpreting photographic images taken from a helicopter, aerial photographs and topographic maps (1:50,000), we estimated overall stream length (m) and change in elevation over this length. Bankfull width and depth (m) were measured along 3–5 randomly chosen transects across each study site. Substrate composition (%) was estimated visually at these transects, with substrates classified as clay and silt (<0.0625 mm), sand (0.0625–2 mm), small gravel (2–32 mm), large gravel (32–64 mm), cobble (64–256 mm), and boulder (>256 mm).

To maximize consistency in visual estimates of substrate composition, the same person made all the estimates.

Epilithon, Bryophytes, and Macrophytes

Biomass of epilithon was estimated by scraping a 4.9 cm² area from the upper surface of one randomly selected stone from five riffle and five pool habitats at each stream site. Because preliminary observations suggested considerably lower epilithic biomass in the constructed stream, we often pooled scrapings from two stones in the same habitat. Stone scrapings were removed using a knife and then placed into plastic vials and frozen for subsequent analysis. In the laboratory, macroinvertebrates were removed from the scrapings, which were then filtered on pre-weighed glass filters, dried at 40°C, weighed, ashed at 550°C for at least 4 h, then reweighed (± 0.1 mg) to calculate ashfree dry mass (AFDM). This was then converted to an areal basis (mg · cm⁻²) and averaged across pools and riffles for a given stream-year.

We estimated visually the coverage by bryophytes and macrophytes of the stream bed within 3–5 0.1 m² areas along each of 3–5 transects in riffle and pool habitats, calculated mean coverage as percentage of bryophytes and macrophytes along each transect, and then pooled these data to create an overall average for each habitat, and then for each stream-year.

Streambank Vegetation, CPOM, and Woody Debris

We characterized stream bank vegetation by quantifying shrub stems located within 1 m of both banks of a stream site and subsequently converted this into stem density (stems · m⁻²). We also estimated visually the percentage of the stream banks covered by herbaceous vegetation (grasses and sedges) within a 1 m belt transect along the stream site.

Replicate samples of CPOM were taken from randomly chosen locations five riffles and five pools in each stream by inserting a cylindrical plastic jar (jar depth and diameter = 9 cm × 8 cm) vertically into the substratum and carefully removing the sample jar without losing its contents. Samples were refrigerated for up to 48 h. In the laboratory, samples were wet washed through 1 mm screening to separate the CPOM from smaller size fractions. We then transferred the material into a plastic tray to allow removal of macroinvertebrates. Samples were then dried at 40°C until their dry mass stabilized.

Volumes of woody debris (≥ 1.0 cm diameter) were determined by counting all pieces longer than 10 cm within each site. We measured the length and diameter at each end

of all pieces and calculated volumes, assuming that woody debris approximated a cylinder. The concentration of woody debris within a site, standardized as $\text{cm}^3 \cdot 100 \text{ m}^{-2}$, was determined by dividing the total volume of woody debris in the site by its surface area.

Leaf Decomposition

We compared rates of organic matter processing in the constructed vs. natural streams by incubating 6 to 8 leaf packs per site in four natural streams (Polar-Vulture A, Polar-Vulture B, Grizzly, Pigeon) and at four sites in the constructed stream. Leaf packs (10 cm $\times$ 15 cm) were composed of plastic mesh bags (mesh dimensions = 2 $\times$ 3 mm) that were filled with 3 g ($\pm$ 0.001 g) of air dried willow (*Salix* spp.) leaves. Leaf packs were deployed in several low velocity riffle habitats per stream by securing packs to metal poles anchored to the river bed. All packs were deployed on 13 June 2000 and retrieved 30–31 days later. Leaf packs were carefully removed from the stream bed and placed into individual plastic bags to minimize loss of leaf matter. Samples were kept cool and in the dark prior to processing within 24 h.

In the laboratory, samples were removed from plastic bags and placed into 1 L pans of water to allow removal of macroinvertebrates and inorganic matter. Samples were then swirled in a 2 L beaker to further separate dense inorganic matter from lighter organic matter and the top portion of the suspension (containing organic matter) was poured through a 70 μm metal sieve. Any remaining organic matter in the bottom of the beaker was removed using forceps. Organic matter was dried at 40°C for 4 days and subsequently weighed to the nearest 0.001 g to calculate the biomass of leaf matter remaining after in-stream incubation. Results from separate leaf packs were averaged to provide one value per stream ($n = 4$ natural streams, $n = 1$ constructed stream).

Benthic Macroinvertebrates

Benthic macroinvertebrates were collected from ten randomly selected locations, one each from five pools and five riffles per stream study site. Samples were taken using a 0.093 m^2 Surber sampler fitted with a 250 μm net. Samples were collected by placing the sampler on the stream bed and agitating the bed materials to a depth of about 5 cm. All samples were preserved in 70% ethanol in the field. Macroinvertebrates were subsequently identified to genus, with the exception of Nematoda, Turbellaria, and terrestrial invertebrates, which were typically identified to Family or Order. Following enumeration, samples were dried to

constant mass at 40°C and weighed ($\pm$ 0.1 mg) to estimate dry mass. Lastly, we quantified functional feeding groups using designations by Merritt and Cummins (1996).

Statistical Analyses

Ordination Analyses

We used Principal Components Analysis (PCA) on the correlation matrix (McCune and Grace 2002) to summarize patterns in physical attributes of the constructed and 10 natural streams. Of the eight stream characteristics included (Table 1), all except discharge were $\log(x)$ or $\log(x + 1)$ transformed. To evaluate the number of interpretable dimensions, a randomization test was conducted in which values were shuffled within variables, then the correlation matrix and eigenvalues were recomputed. This was repeated 999 times, and the actual eigenvalues were then compared with the eigenvalues from the randomizations (McCune and Mefford 1999). To help interpret the ordination and the relationship of the constructed stream to the natural streams, we also conducted hierarchical clustering (centroid linkage, Sorensen's distance measure), overlaying the clustering results onto the PCA ordination graph.

To summarize spatial and temporal patterns in benthic macroinvertebrate assemblages of the constructed and three reference streams, using data collected in 1998–2000, and to compare constructed and natural stream assemblages, we applied non-metric multidimensional scaling (NMS) separately to $\log(x + 1)$ abundance data from riffle and pool habitats. Invertebrates were grouped by families (except as noted above) and taxa that we considered largely lake- or terrestrial-derived, as well as those that were infrequently present (in 3 or fewer stream-years) were dropped from the analyses. This left 26 (riffles) and 23 (pools) taxa in the analyses. NMS used Sorensen's (Bray–Curtis) distance measure and 50 runs of real data employing a random starting coordinate. Final dimensionality was assessed via a randomization test, determining the percentage of 200 randomized runs having stress values equal to or less than the observed stress (McCune and Meford 1999). Varimax and graphical rotations were employed to improve ease of interpretation of the final ordination and the correspondence between the riffle and pool ordinations (McCune and Grace 2002).

One Sample Hypothesis Tests

For the remainder of analyses, we compared habitat and biotic features of the constructed vs. natural streams using a one-sample hypothesis (OSH) test (Sokal and Rohlf 1995). This analysis uses a t -test to determine the

Table 1 Physical characteristics of ten natural streams and the constructed stream in the Barrenlands region of the Northwest Territories, Canada

Stream	Length (m)	Catchment area (km ²)	Rise (m)	Bankfull width (m)	Bankfull depth (m)	Discharge m ³ · s ⁻¹ (Q)	Fines %	Coarse %
Counts	1200	4.3	4	3.7	0.31	0.04	48	33
Grizzly	180	6.5	1	2.9	0.25	0.05	18	67
Polar-Vulture A	200	7.2	4	2.7	0.38	0.05	9	62
Polar-Vulture B	500	6.0	6	2.6	0.49	0.05	2	67
Pigeon	2900	9.0	15	2.4	0.46	0.05	30	58
Pikejaw	1710	38.7	30	3.9	0.52	0.20	10	76
Ursula	250	86.3	20	15.5	0.47	0.50	4	85
Slipper-Rene	410	175.0	5	17.5	0.39	2.00	2	93
Slipper-Lac de Gras	280	185.0	8	20	0.29	2.10	5	80
Norm's Camp	600	222.9	12	35	0.89	2.80	5	73
Average (S.E.)	914 (897)	74.0 (27.6)	10.5 (2.8)	11.5 (11.3)	0.4 (0.2)	0.78 (0.34)	16 (15)	69 (18)
Constructed stream	3400	27.5	14	7.2	0.7	1.2	44	32

Fine substrates include clay, silt, and sand, whereas coarse substrata include cobble and boulder

probability that a single observation (i.e., an overall average measure from the constructed stream) could have originated from a distribution of data points derived from *n* natural streams (using one overall average measure per stream or stream-year). Bonferroni adjusted *p*-values ($p < 0.017$) were used when comparing the natural stream distribution to the three years of constructed stream means. We quantified water depth and velocity in pools and riffles to ensure that samples from the constructed stream were taken from local habitats similar to those in natural streams. We used the Kolmogorov-Smirnov test to examine data for normality and Levene's median test for homogeneity of variances, and transformed data appropriately, where required.

Results

Variation in Characteristics of the Constructed Stream and Natural Streams

A Principal Components Analysis of eight physical variables (Table 1), combined with a cluster analysis, divided the ten natural streams into two broad groups along the first component (47.6% of total variance), which was primarily associated with gradients of catchment area, stream width and discharge, and predominant substrate type to a lesser extent (Fig. 2). Six natural streams, together with the constructed stream, were characterized by smaller catchments, smaller discharges, smaller widths and fine substrate (Fig. 2, Table 1). The second component (30.5% of total variance) was associated with gradients of stream length, depth, and rise (change in elevation). The constructed

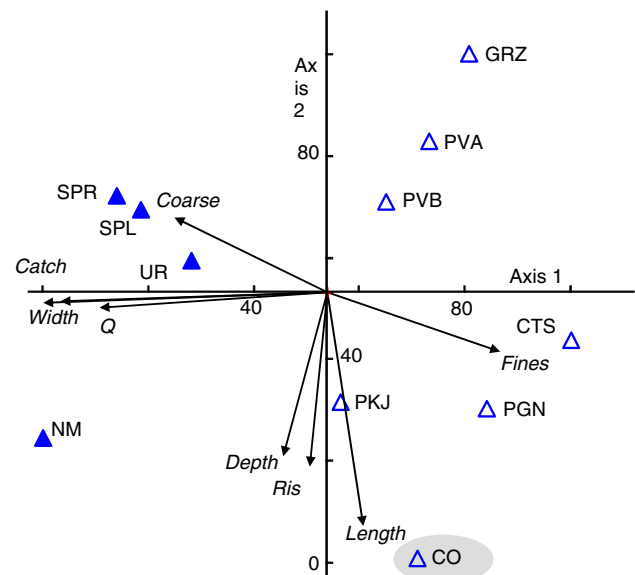
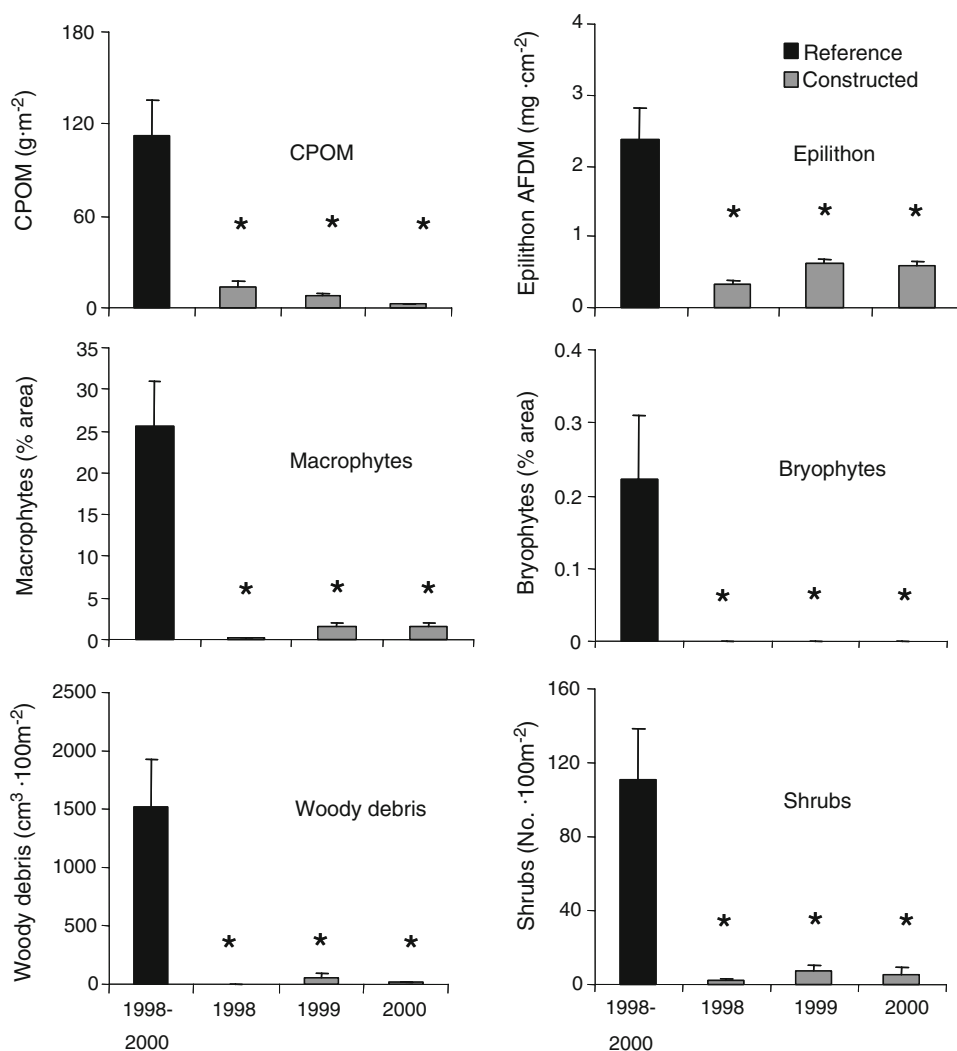


Fig. 2 Principal Components Analysis of physical attributes of 10 natural streams and the constructed stream in the Barrenlands region of the Northwest Territories, Canada. Open and closed triangles indicate membership in two groups of physically similar streams based on a cluster analysis. Vectors (arrows) point in the direction of increasing values for the respective stream attributes, with vector length reflecting the strength of the relationship. Stream site abbreviations are Counts (CTS), Grizzly (GRZ), Polar-Vulture A (PVA), Polar-Vulture B (PVB), Pigeon (PGN), Pikejaw (PKJ), Ursula (URS), Slipper-Rene (SPR), Slipper-Lac de Gras (SPL), Norm's Camp (NMC). See Table 1 for actual data

stream was at the most distal position along this gradient (Fig. 2), due primarily to its exceptional length (Table 1). A randomization test indicated that only two axes were meaningful.

Fig. 3 Mean ($\pm$ standard error) density of shrubs, coverage of aquatic macrophytes, and bryophytes, and biomass of woody debris, coarse particulate organic matter (CPOM), and epilithon from reference streams (1998–2000; black fill) and the constructed stream (each year separately; grey fill) in the Barrenlands region of the Northwest Territories, Canada. AFDM = ash-free dry mass. Asterisks indicate significant differences (Bonferroni adjusted $p < 0.017$; one-sample hypothesis tests) between constructed and natural streams



Based on these results, and the more restricted availability of data from >1 year, we limited our comparisons of biological characteristics of the constructed stream to a subset of 5 natural streams (Pikejaw, Polar-Vulture A, Polar-Vulture B, Grizzly and Pigeon) from the smaller, narrower, fine-substrate group (Fig. 2) with which the constructed stream was clustered (there were data from only one year for Counts, the sixth natural stream in the same group). Furthermore, samples of CPOM, epilithon, and benthic macroinvertebrates were collected from individual pools and riffles having similar ranges in water depth and velocity irrespective of whether they were taken from the constructed or these 5 natural streams.

Streambank Vegetation, CPOM, Woody Debris, Bryophytes, and Macrophytes

One sample hypothesis tests (Bonferroni adjusted $p < 0.017$) revealed marked differences in vegetation-based biological characteristics of the constructed and

reference streams Pikejaw, Polar-Vulture A, Polar-Vulture B, Grizzly and Pigeon (Fig. 3). Biomass of CPOM, epilithon, and woody debris, and the percentage of the streambeds covered by macrophytes and bryophytes were significantly lower in the constructed stream than in natural streams (OSH, Bonferroni adjusted $p < 0.017$, Fig. 3). Density of riparian shrubs, primarily dwarf willow, was also less (OSH, Bonferroni adjusted $p < 0.017$) than that for natural streams throughout the three year study period. Epilithic biomass, extent of macrophyte cover, and density of shrubs within the riparian zone appeared to increase modestly over time in the constructed stream, whereas the coverage of bryophytes, biomass of woody debris, and CPOM either decreased, or displayed no discernable trend over the three year study period (Fig. 3).

Leaf Decomposition

Leaf packs incubated in reference streams (Polar-Vulture A, Polar-Vulture B, Grizzly, Pigeon) retained significantly

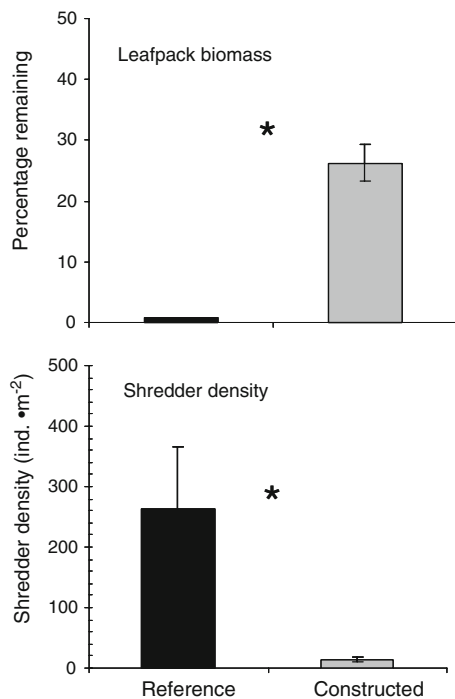


Fig. 4 Mean ($\pm$ standard error) percentage of leaf biomass remaining after 1-month incubation and the density of invertebrate shredders from four sites in the constructed stream and four reference streams in the Barrenlands region of the Northwest Territories, Canada. Invertebrate taxa identified as shredders were based on Merritt and Cummings (1996). Asterisks indicate significant differences (Bonferroni adjusted $p < 0.017$; one-sample hypothesis tests) between constructed and natural streams

less leaf biomass (mean $\pm$ 1 SE = $0.75 \pm 0.1\%$ of original mass) compared to those incubated for an equivalent period in the constructed stream ($26.3 \pm 3.1\%$, (OSH, Bonferroni adjusted $p < 0.017$) (Fig. 4a). The difference in loss of leaf matter between the constructed stream and the natural streams was accompanied by a dramatic difference in the abundance of shredders. Shredders were almost 19 times more abundant in the natural streams (263 ± 102.4 ind/m²) compared to the constructed stream (14 ± 2.9 ind/m², OSH, Bonferroni adjusted $p < 0.017$) (Fig. 4b).

Temporal Variation in Benthic Macroinvertebrate Assemblages

NMS ordination summarized patterns of macroinvertebrate assemblages in both riffle and pool habitats of the constructed stream, as well as in three natural streams, Pikejaw, Polar-Vulture A, Polar-Vulture B, that were in the cluster of physically more similar streams (see Fig. 2) and for which we had data from all three years. For both ordinations, randomization tests recommended using 2 axes. These two axes explained 90.1% and 86.1% of the variation in the original, unreduced data space for riffles

and pools, respectively. Stress values for the two-dimensional ordinations were 8.9 for riffles and 11.7 for pools.

Pool and riffle assemblages in all three natural streams showed strong, consistent patterns associated with year (Fig. 5 upper), in which more centrally located assemblages from 2000 were flanked on either side by those from 1998 and 1999. In the joint plots associated with the two ordinations, taxa with the strongest relationships to the ordination axes (r^2 cutoff = 0.5; McCune and Mefford 1999) mostly reflected patterns of increasing abundance in the direction of the 1998 assemblages (Fig. 5 lower). Taxa that reflected this pattern in both riffles and pools included Lumbriculidae, Turbellaria, Hydroptilidae, Tipulidae, and Chironomidae. Compared to 1999, for example, their densities were 2.4–52.2 times greater in 1998; in some cases, they were completely absent from 1999 samples. More broadly, densities of Plecoptera and Trichoptera declined by 8–28 times between 1998 and 1999. Only a few taxa increased in the direction of the 1999 assemblages (Fig. 5 lower).

In all three years and for both riffles and pools, invertebrate assemblages in the constructed stream fell outside of the triangle of points that connected the three natural stream assemblages (Fig. 5 upper). Densities of numerically dominant taxa in the constructed stream were typically lower than in reference streams (Table 2). Differences between constructed and natural streams were most marked in 1998 when, for example, densities of Lumbriculidae, Turbellaria, Hydroptilidae, Tipulidae, and Mollusca in the constructed stream, if present at all, were only 2–53% of densities in the natural streams. By 2000, however, the densities of these taxa in the constructed stream were 13–147% of those in the natural streams. In the ordination diagrams, the distance of the constructed stream to the natural streams' triangle in 2000 was only 19–32% of the distance in 1998 (Fig. 5).

Mean taxonomic (families) richness in natural streams ranged from 16 to 21 and did not vary appreciably between riffles and pools or among years (Fig. 6). In contrast, mean taxonomic richness in the constructed stream was lower (OSH, Bonferroni adjusted $p < 0.017$) than in natural streams for both habitats and in all three years. Taxonomic richness in the constructed stream increased, however, over the 3-y study period; increases were proportionately larger in riffles than in pools (Fig. 6).

Discussion

Our study revealed significant differences in biological attributes of the constructed stream compared to a suite of nearby, physically similar natural streams during the first three years after construction. In general, stream banks

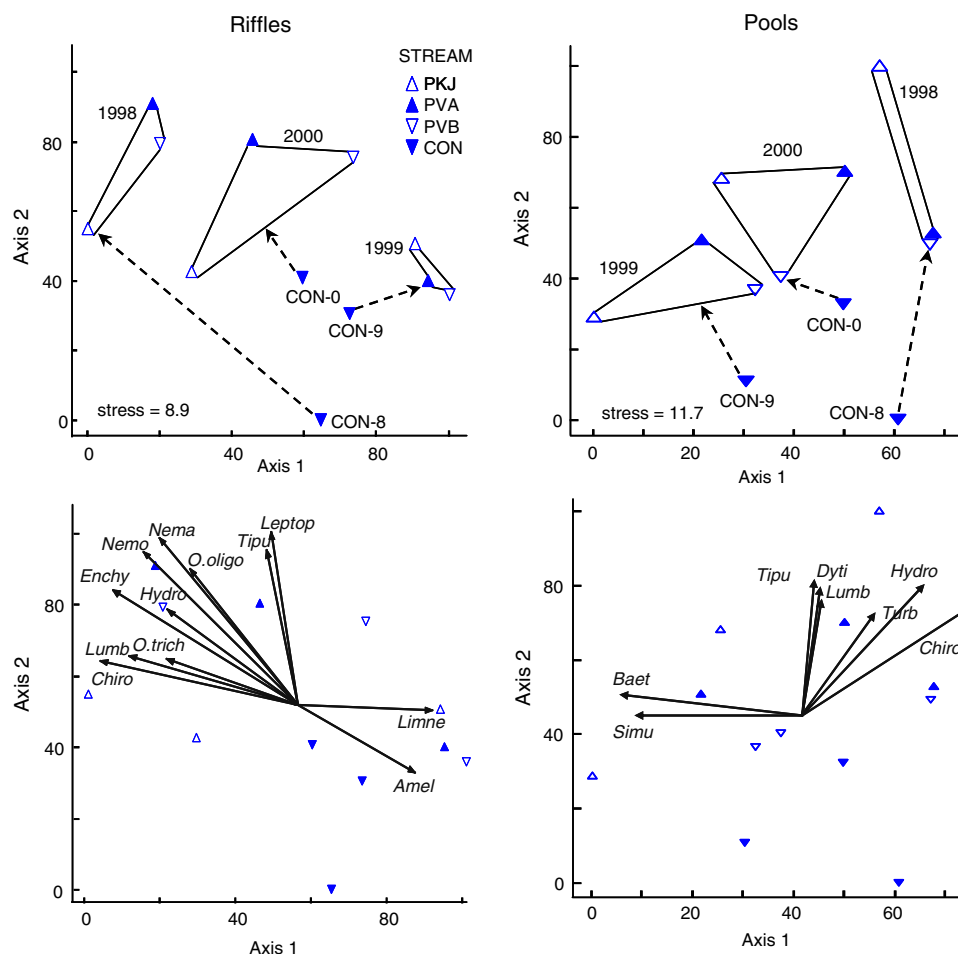


Fig. 5 Nonmetric multidimensional scaling ordinations of benthic invertebrate assemblages from the constructed and three reference streams sampled in 1998, 1999, and 2000 in riffles and pools. Top: Solid lines connect streams within years to delimit the reference stream “space” for each year. Dashed arrows indicate the distance (dissimilarity) of the constructed stream’s assemblage to the appropriate reference “space”. Constructed stream = CON, Polar-Vulture A = PVA, Polar-Vulture B = PVB, Pike-Jaw = PKJ. Numbers after CON indicate sampling year 1998 (8), 1999 (9), and 2000 (0). Bottom: Joint plot vectors radiating from the ordination’s centroid

show the relationship between a taxon’s abundance and the ordination axes; the angle and length of a vector indicates the direction and strength of the relationship, respectively. Only taxa with an $r^2 \geq 0.5$ are shown: Ameletidae (Amel), Baetidae (Baet), Chironomidae (Chiro), Dytiscidae (Dyti), Enchytraeidae (Enchy), Hydroptilidae (Hydro), Leptophlebiidae (Leptop), Limnephilidae (Limne), Lumbri culidae (Lumb), Nematoda (Nema), Nematouridae (Nemo), “other” Oligochaeta (O. oligo), “other” Trichoptera (O. trich), Simuliidae (Simu), Tipulidae (Tipu)

remained largely devoid of vegetation and accrual of epilithon, macrophytes, bryophytes, woody debris, and CPOM was, at best, limited during this period. The constructed stream also processed leaf matter at a significantly lower rate than in natural streams, due, apparently, to a dearth of macroinvertebrate shredders. Although macroinvertebrate assemblages in the constructed stream remained different from natural stream assemblages, taxonomic richness and assemblage composition improved over the 3-year study.

The ability of the constructed stream to develop a macroinvertebrate assemblage similar to those of nearby natural streams will be influenced by the extent to which the constructed stream can be colonized by the regional

fauna and provide a suitable range of environmental conditions, including appropriate food resources, to support the colonists. Models of stream colonization, therefore, could be used to gain insights on the length of time required for macroinvertebrate assemblages in the constructed stream to converge on those in natural streams (e.g., Mackay 1992; Milner and others 2000). Colonization and establishment of a biological community in the constructed stream may approximate the recovery from a large-scale, yet relatively local disturbance, for example, a catastrophic debris flow (Lamberti and others 1991) or flood (Scrimgeour and others 1988). Alternatively, the absence of upstream reaches and the lack of large tributaries suggest that newly created stream habitats, such as those that accompany the

Table 2 Mean densities (ind. · m⁻²) of benthic taxa in the constructed stream and five reference streams in the Northwest Territories, Canada

Habitat	Taxonomic group	Natural streams 1998–2000	Constructed stream		
			1998	1999	2000
Riffles	Nematoda	77.6 ± 19.7	8.8 ± 4.7	24.3 ± 16.6	20.4 ± 16.1
	Oligochaeta	113.5 ± 47.8	13.0 ± 6.9	9.5 ± 6.8	54.0 ± 12.9
	Ephemeroptera	41.2 ± 10.8	15.0 ± 12.8	3.93 ± 2.1	6.9 ± 3.3
	Plecoptera	20.5 ± 13.5	1.5 ± 0.7	0 ± 0	0.1 ± 0.1
	Hemiptera	0.6 ± 0.2	0.1 ± 0.1	0.2 ± 0.2	0.1 ± 0.1
	Trichoptera	14.1 ± 5.3	1.2 ± 0.6	0.9 ± 0.7	0.2 ± 0.1
	Lepidoptera	0.1 ± 0.0	0.1 ± 0.0	0 ± 0	0 ± 0
	Coleoptera	1.00 ± 0.5	0 ± 0	0.4 ± 0.3	0.5 ± 0.4
	Diptera	189.7 ± 76.6	25.7 ± 18.0	67.6 ± 59.4	21.2 ± 9.5
	Empididae	1.3 ± 0.3	3.5 ± 2.2	0.3 ± 0.2	0.3 ± 0.2
	Simuliidae	178.0 ± 77.3	19.7 ± 18.3	65.6 ± 59.7	15.8 ± 9.9
	Tipulidae	10.0 ± 4.5	1.0 ± 0.6	1.5 ± 1.2	4.8 ± 4.1
	Chironomidae	812.4 ± 177.3	1475.4 ± 1006.2	345.3 ± 267.9	446.5 ± 316.8
	Mollusca	22.6 ± 19.3	0 ± 0	0 ± 0	0 ± 0
	Number of (<i>p</i> < 0.05) significant differences		11	10	9
Pools	Nematoda	155.8 ± 35.4	7.7 ± 3.2	22.4 ± 17.2	43.6 ± 17.6
	Oligochaeta	48.6 ± 12.7	9.7 ± 6.3	12.4 ± 5.7	66.6 ± 40.3
	Ephemeroptera	16.6 ± 4.3	0.1 ± 0.0	1.5 ± 0.7	3.2 ± 2.5
	Plecoptera	4.9 ± 2.0	0 ± 0	0.1 ± 0.0	0 ± 0
	Hemiptera	0.46 ± 0.2	0 ± 0	0.1 ± 0.0	0 ± 0
	Trichoptera	9.3 ± 2.8	1.1 ± 0.9	0.1 ± 0.1	0.5 ± 0.3
	Lepidoptera	0.1 ± 0.1	0 ± 0	0 ± 0	0 ± 0
	Coleoptera	1.8 ± 1.1	0.1 ± 0.0	0.4 ± 0.2	0.3 ± 0.1
	Diptera	137.3 ± 110.2	1.8 ± 0.7	3.9 ± 1.6	4.5 ± 1.3
	Empididae	1.5 ± 0.3	0.6 ± 0.4	0 ± 0	0.1 ± 0.1
	Simuliidae	119.6 ± 110.8	0 ± 0	2.3 ± 1.5	0.7 ± 0.3
	Tipulidae	15.6 ± 6.7	1.2 ± 0.6	1.3 ± 0.6	3.2 ± 0.9
	Chironomidae	652.6 ± 139.3	1246.7 ± 468.7	365.5 ± 227.4	603.9 ± 133.7
	Mollusca	19.3 ± 10.8	0 ± 0	0 ± 0	0 ± 0
	Number of significant (<i>p</i> < 0.05) differences		13	10	9

Differences between the reference (3-yr average) and constructed streams (each year separately) were evaluated using one-sample hypothesis tests. Significant (*p* < 0.05) differences are highlighted in bold

retreat of glaciers (Flory and Milner 1999) could provide more appropriate models.

Similar to our study, Flory and Milner (1999) reported high abundances of Chironomidae and a scarcity of other insect (e.g., Plecoptera, Trichoptera, Hemiptera) and non-insect (e.g., Oligochaeta) taxa in the first 3 years of Wolf Creek's appearance following glacial retreat. The initial predominance of larval Chironomidae in the constructed stream is thus not surprising given their high dispersal capabilities that allow for rapid colonization and relatively short generation times that foster a strong numerical response (see Mackay 1992). An abundance of fine particulate organic matter from headwater lakes (Richardson and

Mackay 1991) may also support filter-feeding and collector-gatherer invertebrates in the constructed stream (Jones and others 2003a; Jones and Tonn 2004). Given the similarities between our results and those of Flory and Milner (1999), it is sobering to note that the latter authors developed a conceptual model of community development that describes changes over decades to hundreds of years (O'Neill 1976; Reynolds and Tenhunen 1996). In the absence of active remediation measures, therefore, we suggest that the biological community in the constructed stream will require a decade or more before it approximates communities in natural streams. Although ordinations suggested that macroinvertebrate assemblage composition in the

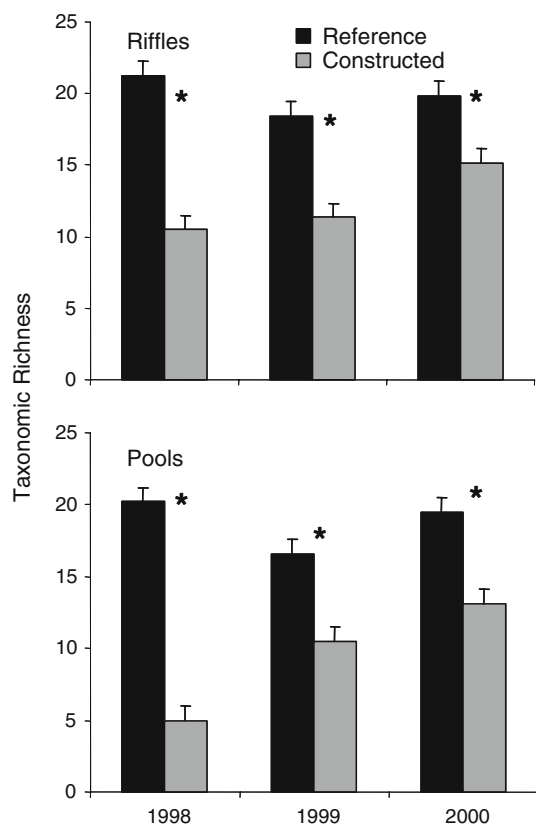


Fig. 6 Mean ($\pm$ standard error) site taxa richness (family) of macroinvertebrates in riffle and pool habitats of the constructed and reference streams, Pikejaw, Polar-Vulture A, Polar-Vulture B, Grizzly and Pigeon, during 1998–2000. Asterisks indicate significant differences (Bonferroni adjusted $p < 0.017$; one-sample hypothesis tests) between the constructed and the reference streams. Bonferroni adjusted tests also indicated higher taxonomic richness in the constructed stream during 2000 than in 1998 and 1999 in riffles, and increasing taxonomic richness in each of the three successive years for pools in the constructed stream

constructed stream was approaching those in the natural streams, these analyses may have exaggerated the movement towards convergence due to the family-level taxonomy used, and the fact that rare taxa were not included. They do indicate, however, that colonization of some groups is occurring at rates faster than those suggested by Flory and Milner (1999). Although some community measures appear to be approaching the natural streams during the first 3 years, the restoration trajectory need not be linear (Choi 2007). The rate of change, e.g., addition of species, may slow (saturation curve). “Supertramp” species are likely the first to colonize followed by species with poorer dispersal and colonization abilities. Other, more ecosystem-level measures such as macrophytes and woody debris (Fig. 3), showed no such recovery trend. These other ecosystem measures likely mediate the recovery of benthos such that full recovery is dependent on succession of the riparian zone and the production of organic matter.

Factors other than colonist availability may be more limiting for macroinvertebrates. At the whole-stream scale, water quality and nutrients between the constructed and natural streams were similar (Jones and others 2003b); prominent physical habitat differences included a lack of large substrata (Jones and others 2003b). We suggest, however, that a scarcity of coarse organic matter is a major contributor to the low densities of macroinvertebrates inhabiting the constructed stream following its development (see also Malmqvist and others 1991; Laasonen and others 1998). Accrual of epilithic biomass, for example, was low in the constructed stream over the 3-year period and likely reflected low rates of colonization, combined with low water temperatures, low concentrations of nutrients, and high turbidity in the spring and scouring by sediment (Jones and others 2003b). Furthermore, although the natural stream macroinvertebrates displayed a strong year-to-year pattern of assemblage structure and abundance, reflecting the warm, dry conditions of 1998 and the cool, wet conditions of 1999 (Jones and others 2003b), the macroinvertebrates in the constructed stream appeared more limited in their abilities to respond to weather-related patterns in productivity. As well, the biomass of CPOM actually declined in the constructed stream from about 15 g/m² in 1998 to 5 g/m² in 2000, suggesting that the construction of the stream resulted in an initial pulse of CPOM, followed by a period of utilization and/or poor retention that subsequently reduced the availability of these materials to support shredders.

Consistent with other studies, lower leaf processing rates in the constructed stream were accompanied by significantly lower densities of shredders compared to those at natural stream sites (Wallace and others 1997; Sponseller and Benfield 2001; Johnson and others 2003). Larval *Tipula* (Tipulidae, Diptera) and Trichoptera were the numerically dominant shredders in natural streams, but were present in low densities at leaf-pack sites in the constructed stream, and large particle detritivores were largely absent from the entire constructed stream.

Low macroinvertebrate abundances and low coverage by macrophytes and bryophytes in the constructed stream may also have resulted because much of it was still in a phase of hydraulic flux, with marked erosion and deposition of river bed materials. Our observations suggested that these processes were continuing to alter the stream profile. As well, the spring freshet is accompanied by active bed movement, sediment mobilization, bed scour, and high water turbidity (Jones and others 2003b), which are known to hamper macroinvertebrate establishment and growth (Waters 1995; Shaw and Richardson 2001). The virtual lack of local riparian vegetation to serve as a source pool for large woody debris and CPOM, combined with low retention of large woody materials, also precludes the

establishment of snag-dwelling invertebrate communities (Benke and others 2001) that reside in natural streams. In turn, lower macroinvertebrate densities resulted in reduced forage base and growth of young-of-the-year Arctic grayling in the constructed stream (Jones and others 2003a, b).

Conclusions

Assessments of environmental degradation and restoration need to include evaluations of multiple biological attributes, including algal, benthic macroinvertebrate and fish assemblages (e.g., Sawyer and others 2004; Hering and others 2006, Wang and others 2007). Our analyses indicate that the productive capacity of the constructed stream three years after its construction was lower than that of natural streams in the region and that additional time or active compensation measures would be required. The productive capacity of the constructed stream could be enhanced through a diversity of means. The stream was engineered to provide a variety of slope profiles to create pool, riffle and run habitats (Jones and others 2003b), and included active measures of introducing some boulders within the channel to create hydrologically complex flow features. Biological interventions, however, e.g., the planting of riparian and instream vegetation, were minimal during the first three years following construction of the stream and fraught by hare and caribou browsing. Our data suggest that lower densities and dissimilar composition of macroinvertebrate assemblages of the constructed stream were likely related to stream and riparian instability and especially a paucity of autochthonous and allochthonous organic matter. If correct, then significantly more large cobble and boulders and the addition of woody debris and CPOM, combined with planting of shrubs and herbs along the stream, could provide a source of allochthonous matter while improving the physical stability of stream system required to keep the vegetation and CPOM in the system. Furthermore, the introduction of macrophytes and large substrata from natural streams that support dense bryophyte communities could increase primary production and provide sources of propagules and organic matter to the stream. These measures should also be accompanied by a rigorous monitoring program that measures structural and functional attributes of the stream ecosystem beyond the number and size of Arctic grayling, such as plant and invertebrate communities and organic matter accrual and retention. This approach can provide further insights on the causes of the low performance of the constructed stream, guide effective management, and demonstrate when the losses in the productive capacity of destroyed natural streams have been offset. Based on our findings, we suggest that a long-term monitoring program to quantify improvement in the

condition of the constructed stream should focus on macroinvertebrate assemblage structure and function, macro- and micro-vegetation, and organic matter.

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